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Fórmula para la proyección ortogonal en un sistema ortonormal finito

Teorema 1. *Sea H un espacio de Hilbert y $S = \{u_i\}_{i=1}^n$ un sistema ortonormal. Consideramos $M = \operatorname{span}(S)$, entonces*

$$\forall x \in H : P_M(x) = \sum_{j=1}^{n} \langle x, u_j \rangle u_j.$$

***Demostración*:** Por la `prop-subesp-vectorial-generado-cerrado/`**??**, M es cerrado y, por tanto, por el `teo-proyeccion-ortogonal/`Teorema 1, tenemos que $x = P_M x + P_{M^\perp} x$. Entonces, basta demostrar que $x - \displaystyle\sum_{j=1}^{N} \langle x, u_j \rangle u_j \in M^\perp$. Por ser $M = \operatorname{span}(S)$, es suficiente demostrar que $\forall k \in \mathbb{N}_n : \left\langle x - \sum_{j=1}^{N} \langle x, u_j \rangle u_j, u_k \right\rangle = 0$. Pero como $\langle u_j, u_k \rangle = \delta_{jk}$, tenemos que

$$\left\langle x - \sum_{j=1}^{N} \langle x, u_j \rangle u_j, u_k \right\rangle = \langle x, u_k \rangle - \sum_{j=1}^{N} \langle x, u_j \rangle \underbrace{\langle u_j, u_k \rangle}_{=\delta_{jk}} = \langle x, u_k \rangle - \langle x, u_k \rangle = 0.$$

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